Domain Adaptation via a Task-Specific Classifier Framework for Remote Sensing Cross-Scene Classification

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***Abstract*— The scene classification of high spatial resolution (HSR) imagery involves labeling an HSR image with a specific high-level semantic class according to the composition of the semantic objects and their spatial relationships. As such, scene classification has attracted increased attention in recent years, and many different algorithms have now been proposed for the cross-scene classification task. However, the recently proposed scene classification methods based on deep convolutional neural networks (CNNs) still suffer from domain shift problems, because of the training data and validation data not following the assumption of independent and identical distributions. The employment of generative adversarial networks has been found to be an effective way to bridge the domain shift/gap. However, the existing cross-scene classification methods do not use the classification information in the target domain, and the domain classifier is task-independent for different scene classification tasks. In this article, to solve this problem, domain adaptation via a task-specific classifier (DATSNET) framework is proposed for HSR image scene classification. Task-specific classifiers and minimizing and maximizing, " i.e., minimaxing," of the classifier discrepancy are integrated in the DATSNET framework. The task-specific classifiers are proposed to align the distributions of the source domain features and target domain features by utilizing task-specific decision boundaries in the target domain. In order to align the two task-specific classifiers' feature distributions, minimaxing the defined discrepancy between the different classifiers in an adversarial manner is proposed to obtain better task-specific classifier boundaries in the target domain and a better-aligned feature distribution in both domains. The experimental results obtained with different remote sensing cross-scene classification tasks demonstrate that the proposed method achieves a significantly improved performance compared with the other state-of-the-art remote sensing cross-scene classification algorithms.**

***Index Terms*— Convolutional neural network (CNN), discrepancy, domain adaptation, scene classification, task specific.**

NOMENCLATURE

$\boldsymbol{C}$	Cost matrix of transporting between probability distributions.
$\boldsymbol{P}$	All the possible transportation plans.
$p_i(x)$	ith classifier output trained on the source domain.
X_s	Labeled source domain training samples.
X_T	Unlabeled source domain training samples.
F	Task-specific classifier.
G	Feature generator.
$II[k = y_s]$	Indicator function of whether y_s equals class k. If yes, the value is 1; otherwise, the value is 0.

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I. INTRODUCTION

WITH the rapid development of Earth observation technologies, more and more very high-resolution (VHR) images are becoming available to both the public and researchers. Because of the improvement of the spatial resolution of remote sensing images, VHR images provide very precise geometrical information, subtle land-use object texture, and a clearer layout of the objects in the high-resolution images. As a result, how to obtain the high-level semantic categories of remote sensing scene images, e.g., residential or education areas, has attracted the attention of many researchers. However, due to the diversity and complicated characteristics of the remote sensing image scenes in VHR images, the remote sensing scene classification task stills remains a challenge.

The traditional scene classification methods can be classified into the following three categories: 1) methods based on semantic objects [1]–[5]; 2) methods based on midlevel feature modeling [6]–[14]; and 3) methods based on deep convolutional neural networks (CNNs) [14]–[22]. For methods based on semantic objects, they generally contain three steps: 1) semantic object extraction; 2) modeling of the relationships between the objects which constitute the scene; and 3) class inference [23]. However, the scene classification accuracy of such methods is heavily reliant on the semantic object extraction accuracy and the expression of the relationships between the semantic objects. For methods based on midlevel feature modeling, the process of midlevel feature modeling involves the following steps: 1) extraction of the low-level features; 2) visual word learning; and 3) modeling of the midlevel features. In the process of extracting the low-level features, color, structure, and texture descriptors, such as the scale-invariant feature transform (SIFT) descriptor [24] and the local binary pattern (LBP) descriptor [25], are used to extract the low-level features from image patches in the original images. Then, in visual word learning, a clustering method, such as the k-means clustering algorithm or a sparse coding

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algorithm, is exploited for dictionary learning. In the final stage, a midlevel feature modeling method, such as the bag of visual words (BoVW) model [6], [26]–[28] or the probabilistic topic model [29], [30], is used to model the semantic relationship between the scene "words" and the scene class, to generate powerful representative features for remote sensing scene classification.

Differing from the above methods based on manually designed features, methods based on CNNs are end-to-end frameworks for automatic multilevel feature extraction and scene class prediction. In the remote sensing field, CNN-based methods have now been successfully applied in objection detection [31]–[34], semantic segmentation [35]–[39], and hyperspectral pixel classification [39]. The methods based on CNNs can be categorized as follows: 1) methods that involve training from scratch; 2) methods that use an ImageNet pretrained network as a feature representation extractor; and 3) methods based on fine-tuning from an ImageNet pretrained network. As an example of a method based on training from scratch, Han *et al.* [40] trained an autoencoder model from scratch using unlabeled data and then fine-tuned the CNN layers using labeled samples. However, the methods based on training from scratch require a large amount of labeled training samples. Fine-tuning from a network pretrained on ImageNet has been proposed to address the dilemma of the lack of training samples. For example, Castelluccio *et al.* [41] initialized a CNN with parameters pretrained on ImageNet, achieving a better classification accuracy than the training from scratch approach. Among the methods based on a pretrained network for remote sensing image feature representation, Hu *et al.* [16] proposed different strategies to utilize the ImageNet pretrained network as a feature extractor for the scene classification task. In addition, Zhu *et al.* [42] proposed an adaptive feature normalization strategy to integrate fully sparse topic modeling and feature representation based on a pretrained network.

The aforementioned methods have shown that pretrained CNNs are effective tools to extract feature representations for different scene classification tasks. The traditional scene classification methods follow the hypothesis that the training data and verification samples are independent and identically distributed. Thus, it is to be expected that the simple fine-tuning strategy will fail to achieve an acceptable performance for cross-scene classification, which is mainly due to the domain shift/gap problem. The reason for this problem is that the training data and validation data do not follow independent and identical distributions. In the remote sensing field, when training data and validation data are collected from different imaging places and times, adopting the same model is likely to fail because of the different data distributions, which are caused by the influence of the sensor, cloud contamination, shadows, and so on.

Recently, domain adaptation methods have been proposed to adapt a trained model by using an existing labeled dataset. As for the scene classification task, a number of deep learning-based domain adaptation methods have been designed for cross-domain classification [43]–[50]. In remote sensing cross-scene classification, the most common phenomenon is that the feature space is different between the source and target domains, and there are no labeled data in the target domain in unsupervised domain adaptation (UDA). Based on these facts and the cross-scene classification methods published in the remote sensing field, the existing methods can be classified into two categories: 1) cross-scene classification methods based on statistics and 2) cross-scene classification methods based on generative adversarial networks (GANs). The methods based on statistics achieve transfer learning by adding a loss term based on statistics-based divergence in the feature layers, such as the maximum mean discrepancy [43]–[48] or CORAL loss [49], to align the feature distributions of the source and target domains. However, such methods are built on a strong hypothesis for the feature distribution, and the statistics criterion is only applied at the feature level, while the classification information in the target domain is not utilized. Methods based on GANs avoid this problem by adding a domain classifier at the top of the feature layer to achieve feature distribution alignment in an adversarial manner [50]. However, the domain classifier only plays the role of judging whether the sample comes from the source domain or not. In summary, the existing cross-scene classification methods do not utilize the classification information in the target domain, and the domain classifier is task-independent for different tasks.

In this article, to utilize the classification information in the target domain and make the cross-scene classification task-specific, the domain adaptation via a task-specific classifier (DATSNET) framework is proposed to address the cross-scene classification task. In DATSNET, the task-specific classifiers and the minimizing and maximizing (i.e., "minimaxing") of the task-specific classifier discrepancy are integrated to generate a more discriminative feature representation for cross-domain remote sensing images. In the task-specific classifier framework, two CNNs are used as the task-specific classifiers to utilize the classification information in the target domain. By minimaxing the defined discrepancy phase, the feature generator and the classifier are optimized to align the feature distributions of both domains. The main contributions of this article are given as follows.

1) *DATSNET Framework:* It is proposed to learn transferrable image features and align the feature distributions between the source domain and target domain remote sensing images. Based on the task-specific classifiers and the adversarial learning approach, the alignment of the feature distributions between the source domain and target domain images is achieved by minimaxing the discrepancy between the two different classifiers, to optimize the networks in an adversarial manner.
2) *Task-Specific Classifiers for UDA:* The task-specific classifiers are proposed to align the feature representation distributions of the source domain features and target domain features by exploiting task-specific class boundaries. Differing from the previous methods, the proposed method utilizes two task-specific classifiers to improve the alignment of the distributions of the different domains through adversarial learning by minimaxing the discrepancy of the two task-specific classifiers.

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The feature spaces of the source domain and target domain are gradually and adaptively aligned in the feature learning process. The key to the task-specific classifiers for UDA lies in the definition of "discrepancy" for the different classifiers. To better capture the discrepancy between the different classifiers, the L1 distance, Kullback–Leibler (KL) divergence, and Wasserstein distance are introduced to measure the outputs of the different classifiers. The comparison of the different discrepancy measures provides an alternative choice for different cross-scene tasks.

3) The different classifiers' feature distributions are effectively aligned by minimaxing the defined discrepancy to optimize the networks in an adversarial manner. In the proposed method, by minimizing and maximizing the defined discrepancy between the different classifiers in an adversarial manner, the feature generator and the classifier are optimized. As a result, better task-specific classifier boundaries in the target domain and a better-aligned feature distribution in both domains can be obtained.

The rest of this article is organized as follows. Section II gives a brief description of the related background to the proposed method. Section III provides the full details of the proposed method for the cross-scene high spatial resolution (HSR) imagery scene classification task. A brief description of the datasets, the experimental results, and an analysis of the experimental results are provided in Section IV. Finally, our conclusions are drawn in Section V.

II. Background

The proposed method is highly related to domain adaptation and GANs. In this section, we give a brief introduction to domain adaptation and domain adaptation based on GANs.

A. Domain Adaptation

The ideal scenario for remote sensing scene classification is that there are enough labeled samples for training a transferable model for different application scenarios. However, in a real-world application, the feature distribution of the source training samples will not be the same as the feature distribution of the test data, which results in the trained model failing [46]. The shift/gap between the source domain and target domain is generally due to the differences in the data acquisition, atmospheric conditions, and the spatial appearance of the geospatial land-use objects. To solve the above problem, transfer learning or domain adaptation methods have been introduced to improve the alignment of the feature distributions between both domains, so as to reuse the trained model.

In the domain adaptation setting, there are two domains, i.e., the source domain and the target domain, which are often collected in different temporal and spatial situations. The domain adaptation methods can be classified into supervised domain adaptation methods and UDA methods. The difference between the two categories is whether the target domain labels are given or not. In the UDA setting, given a labeled source domain $D_s = (\boldsymbol{x}_i, y_i)_{i=1}^{n}$ and unlabeled target domain $D_t = (\boldsymbol{x}_j)_{j=n}^{n+m}$, the source domain distribution $P(\mathbf{x_s})$ and the target distribution $P(\mathbf{x_t})$ will be different but similar. The purpose of domain adaptation is to reuse the knowledge of the source domain and utilize the common information between the source domain and the target domain. For the remote sensing cross-scene classification task, the purpose of domain adaptation is to use the source domain labeled data D_s to learn a classifier $f_t(\boldsymbol{x}_t) = \boldsymbol{y}_t$ to predict the label set $\boldsymbol{y}_t$ for the target domain D_t.

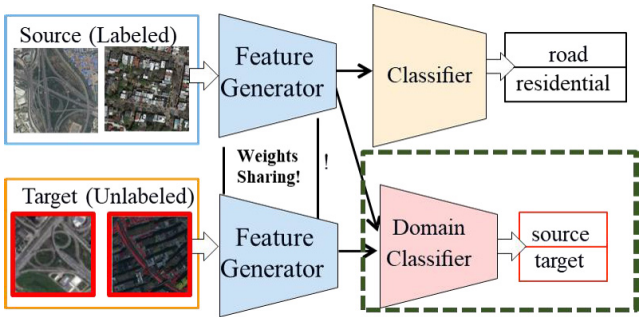


Fig. 1. GAN-based framework for cross-scene classification.

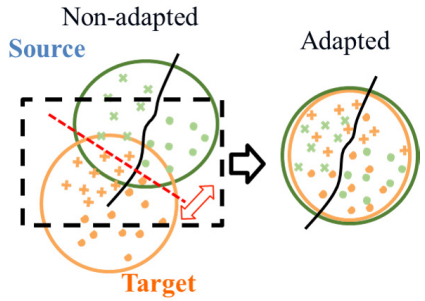


Fig. 2. Feature distribution for the GAN-based framework for cross-scene classification, comparing the nonadapted and adapted distributions.

B. Domain Adaptation Based on GANs

GANs [50] are a kind of network that originated in game theory. The discriminator and the generator are the two main parts of a GAN. The network achieves the final convergence of the network through the confrontation training between the discriminator and the generator. The overall framework is shown in Fig. 1.

In remote sensing domain adaptation tasks, the overall GAN-based domain adaptation framework contains three main parts [46]: 1) the source domain classifier; 2) the feature generator; and 3) the domain classifier. The feature generator is used to generate features to deceive the discriminator. Through the adversarial training between the feature generator and the domain classifier, the feature distributions of the source domain remote sensing images and the target domain remote sensing images can be aligned. As the result, the source classifier trained on the source domain images can be reused to effectively classify the target domain remote sensing images, as shown in Fig. 2. The term "nonadapted" denotes the feature distribution before using the GAN, and the term "adapted" denotes the feature distribution after using the GAN.

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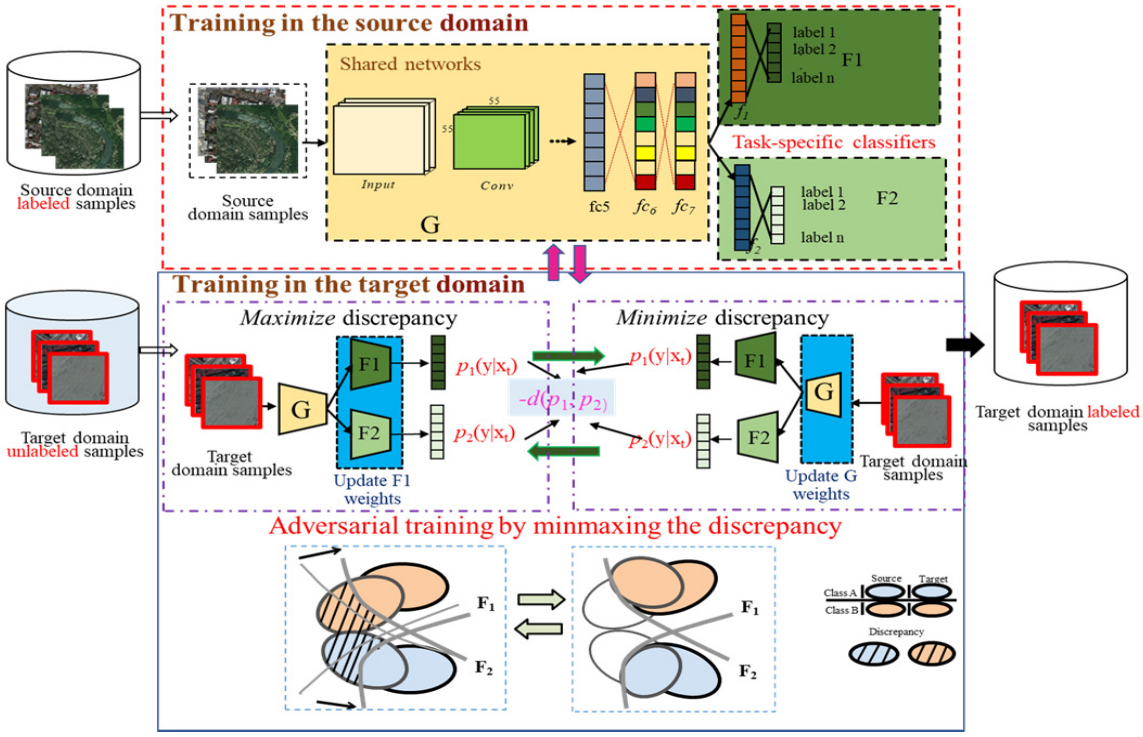


Fig. 3. Flowchart of scene classification based on the proposed framework for remote sensing cross-scene classification.

III. Task-Specific Classifier Framework for Remote Sensing Cross-Scene Classification

In previous studies, classification in the target domain has not been utilized and, as a result, the feature generator may generate vague features near the classification boundaries. Subsequently, the performance of the classifier drops significantly. To effectively utilize the classifier information in the target domain, a DATSNET framework is proposed to solve the problem of remote sensing cross-scene classification. Three tasks need to be addressed: 1) scene/task-specific classifiers for domain adaptation; 2) the task-specific classifier discrepancy; and 3) adversarial training.

A. Scene/Task-Specific Classifiers for Domain Adaptation

In real-world scene classification applications, different target domains can be obtained due to the varying remote sensing imaging areas and conditions. The characteristics of the target domain can vary, including the spectral, geomorphological, and textural characteristics, and the scene category numbers can also vary, which results in different cross-scene classification tasks. Task-specific classifiers mean that the classifiers trained for each specific cross-scene classification task are different. A task-specific classifier means that we have different scene task classifiers for different applications. For example, for an urban scene classification task, the commercial, residential, schools, and other classes need to be classified. For a rural area, the meadow, forest, farmland, and other classes need to be classified. In a real-world application, we can be faced with different domain adaptation classification tasks. For example, transfer learning from Wuhan to Shanghai and transfer learning from New York to Washington. For the different tasks, the classification systems will be different. The classifier for the different scene classification tasks is referred to as a task-specific classifier. The source and target domains share the same category spaces. The use of scene/task-specific classifiers allows us to utilize classification in the target domain, compared with using a domain-specific classifier, and to preserve the characteristics of the specific target domain.

In the proposed framework, two CNNs are used as the task-specific classifiers to improve the alignment of the feature distributions. As shown in Fig. 3, the two task-specific classifiers generate feature representations from the generator using shared networks. The two classifiers output all the classes common to both domains. The source and target domains share the same category space. The two task-specific classifiers attempt to achieve a good classification of the source domain samples using different hyperparameters and are trained to find the samples in the target domain that are near to the classification decision boundaries, whose classification outputs are inconsistent. The main purpose of the proposed framework is to align the feature distributions between the different domains by minimaxing the discrepancy between the two classifiers. For both classifiers, a pretrained CNN based on VGG or ResNet-50 is used as the base network.

B. Scene/Task-Specific Classifier Discrepancy

In the DATSNET framework, how to define the discrepancy between the scene/task-specific classifiers stills remains an open problem. In mathematical statistics, there are several measures that can be used to represent the distance between two different distributions: 1) the Manhattan distance; 2) the KL divergence; and 3) the Wasserstein distance.

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1) Manhattan Distance: The Manhattan distance, which is also called the L1 distance or the L1 norm, is widely used in machine learning research. It is the sum of the absolute difference of the components of the vectors [51]. In the field of optimization, the L1 distance can obtain the sparsest solution. For the HSR image cross-scene classification task, the Manhattan distance can be expressed as follows:

$$D_{l_1}(p_1(x), p_2(x)) = \sum_{i=1}^{n} |p_{1i}(x) - p_{2i}(x)| \tag{1}$$

where $(p_{1i}(x), p_{2i}(x))$ are the vectors $p_1 = (p_{1i}, p_{2i}, \ldots, p_{\text{in}})$ and $p_{2i} = (p_{21}, p_{22}, \ldots, p_{2n})$ considering the classifier output trained on the source domain.

2) KL Divergence: In the probability theory field, the KL divergence is a special case of f-divergence, which is a function that is applied to measure the statistical difference between two different probability distributions. The KL divergence was first introduced in the information processing field in 1951 [52] and is now widely used to characterize the relative entropy in information systems and information gain in the statistical field.

As $p_1(x)$ and $p_2(x)$ denote the softmax probability of the two different task-specific classifiers for cross-scene VHR remote sensing images, the KL divergence of the two different task-specific classifiers is expressed as follows:

$$D_{l_1}(p_1||p_2) = \sum_{x \in X} p(x) \log\left(\frac{p_1(x)}{p_2(x)}\right). \tag{2}$$

3) Wasserstein Distance: The Wasserstein distance, which is also called the Earth mover's distance, is another widely used metric for measuring the distance between probability distributions [53]. The Wasserstein distance considers that the source domain distribution is regarded as a group of Earth in the source domain space, while the target domain distribution is seen as a group of holes in the target domain space. The Wasserstein distance measures the amount of work it takes to fill these target domain holes with the group of source domain Earth. It measures the minimum cost of transporting a probability distribution to another distribution. Due to the consideration of the geometric characteristics between the different distributions, the Wasserstein distance has drawn the attention of many researchers in the machine learning community [53].

If we consider two different probability distributions $p_1(x)$ and $p_2(x)$, the Wasserstein distance can be expressed as follows:

$$L_C = \min_C \langle \boldsymbol{C}, \boldsymbol{P} \rangle \tag{3}$$

$$\text{s.t. } P1 = p_1(x), \quad P^{\mathrm{T}}1 = p_2(x). \tag{4}$$

The equation explains the minimum cost of transporting a probability distribution $p_1(x)$ to another distribution $p_2(x)$, for which the most common choice of cost matrix is the Euclidean distance matrix.

C. Adversarial Training

The last key step of the proposed framework is adversarial training of the two classifiers by minimaxing the defined discrepancy. The training steps of the proposed method are given as follows.

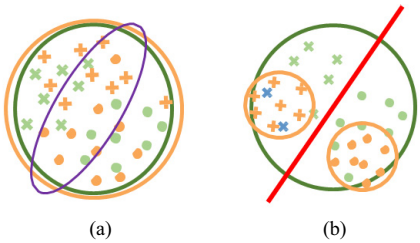


Fig. 4. Task-specific feature distributions in both domains of the proposed framework. (a) GAN-based. (b) Proposed method.

Step A: Both scene classifiers and the scene image feature generator are trained to classify the source domain training samples. The generator and both classifiers are trained during Step A to minimize the cross-entropy loss, in order to obtain the task-specific classifiers and task-specific feature representations. The objective during Step A can be illustrated as follows:

$$L_{\text{cls}}(X_s, Y_s) = -E_{(x_s, y_s)\sim(X_s, Y_s)} \sum_{k=1}^{K} \text{II}[k = y_s] p(y|x_s) \tag{5}$$

$$p(y|x_s) = \log F(G(x_s)). \tag{6}$$

In this step, the loss function should be minimized with respect to G and F.

Step B: In this step, the weight in the scene image feature generator is fixed and the discrepancy is maximized for the fixed scene classifiers on the unlabeled samples in the target domain images. Both classifiers are trained during Step B to maximize the discrepancy loss, to detect the target domain unlabeled samples not supported by the classifier decision boundaries. Classification loss using the source domain samples is added to make the network training process converge, as the network would not converge without adding the source domain loss. The objective during Step B can be illustrated as follows:

$$L_{\text{adv}}(X_s, Y_s) = -L_{\text{cls}}(X_s, Y_s) + L_{\text{dis}}(X_T) \tag{7}$$

$$p_i(\boldsymbol{y}|\boldsymbol{x_t}) = \log F_i(G(\boldsymbol{x_t})) \tag{8}$$

$$L_{\text{dis}}(X_T) = = E_{(\boldsymbol{x_t})\sim(X_T)}\left[d(p_1(\boldsymbol{y}|\boldsymbol{x_t}), p_2(\boldsymbol{y}|\boldsymbol{x_t}))\right]. \tag{9}$$

In this step, the loss function should be maximized with respect to F.

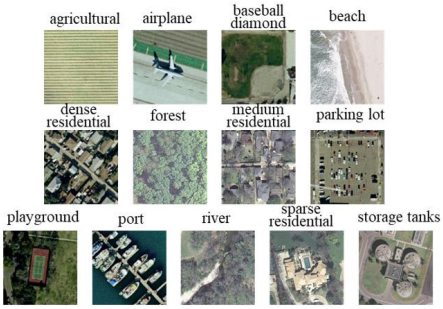
Step C: The scene image feature generator is trained and the discrepancy is minimized for the fixed scene classifiers on the unlabeled samples of the target domain. The weights of both classifiers are then fixed, and the scene feature generator is trained to align the task-specific feature distributions in both domains using the task-specific classifiers. The objective during Step C can be illustrated as follows:

$$L_{\text{adv}}(X_s, Y_s) = L_{\text{cls}}(X_s, Y_s) + L_{\text{dis}}(X_T). \tag{10}$$

In this step, the loss function should be minimized with respect to G.

The classifiers and generator are trained in an adversarial manner to make use of the task-specific information and align the task-specific feature distributions in both domains, as shown in Fig. 4.

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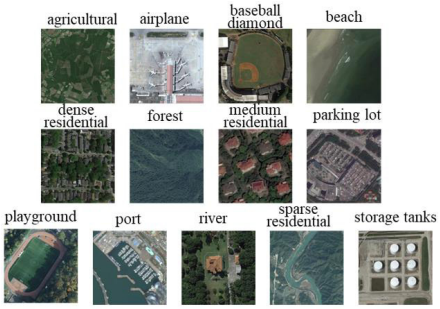


Fig. 5. Samples of the 13 common classes in the AID and UCM datasets.

IV. Experiments and Analysis

A. Experimental Setup

In order to evaluate the effectiveness of the proposed DATSNET framework for the cross-scene classification task, we built six different cross-scene classification tasks with the widely used 21-class UC Merced (UCM) dataset [27], the 30-class AID dataset [54], and the large-scale NWPU-RESISC45 dataset [55]. The six tasks were: from UCM to AID, from AID to UCM, from UCM to NWPU, from NWPU to UCM, from AID to NWPU, and from NWPU to AID. For the UCM dataset, each class has 100 images with a size of 256 × 256 pixels and a spatial resolution of 1 ft. For the AID dataset, each class of the datasets contains 220–420 images with a size of 600 × 600 pixels and a spatial resolution ranging from 0.5 to 0.8 m. For the NWPU dataset, each class contains 700 images with a size of 256 × 256 pixels and a spatial resolution ranging from 0.2 to 30 m. The traditional methods based on statistics and GANs were compared. ResNet-50 and VGG16 networks were used as the backbone networks. As the networks require inputs of the same size, the remote sensing images in the three datasets had to be resized to 256 × 256 to meet the network input requirements. The Adam algorithm [56] was adopted to train the proposed network, with momentum $\beta_1 = 0.9$. The experiments were implemented in the PyTorch machine learning library [57], which was developed by Facebook's AI Research Lab. To evaluate the performance of the proposed DATSNET framework, the experimental results obtained with the traditional domain adaptation methods based on statistics-based divergence are exhibited for comparison, i.e., deep domain confusion (DDC) [58], the domain adaptation network (DAN) [43], and Deep CORAL [49]. Tables VII–XII list the quantitative results for the six tasks when using VGG16 as the backbone network. We also provide the results of the GAN-based domain adaptation method using domain-adversarial training [domain adaptive neural network (DANN)] [44]. The results obtained by the method proposed in the latest article by Ammour *et al.* [47], i.e., the asymmetric adaptation neural network (AAAN), are also presented to compare with the proposed DATSNET framework when transferring from UCM to AID and from AID to UCM. The variants of the proposed method based on the L1 distance, KL divergence, and Wasserstein distance are denoted as DATSNET_L1, DATSNET_KL, and DATSNET_WA, respectively. To test the robustness of the method, we set a set of random seed when using ResNet-50.

TABLE I

Overall Classification Accuracy (%) Comparison for the AID ↔ UCM Experiment Using VGG16 as the Backbone Network

Method	A_U(%)	U_A(%)
Source only	66.34±1.41	61.27±0.67
DDC	66.45±0.425	58.32±0.66
DAN	84.22±1.30	65.02±0.71
Deep CORAL	79.60±1.36	60.71±0.47
DANN	81.88±2.22	63.55±0.32
AANN	74.25	71.70
DATSNET_L1	86.44±0.43	77.12±0.91

TABLE II

Overall Classification Accuracy (%) Comparison for the AID ↔ UCM Experiment Using ResNet-50 as the Backbone Network

Method	A_U(%)	U_A(%)
Source only	62 34±2.26	61.80±0.52
DDC	68.38±0.87	52.84±1.22
DAN	75.40±2.13	56.63±1.24
Deep CORAL	65.69±0.12	53.94±0.15
DANN	67.21±0.42	53.94±0.50
DATSNET_L1	85.67±0.30	88.60±1.42

In the experimental setting, the target domain data were divided into a validation set and a test set with a proportion of 1:4. To ensure that the network architecture was fixed, the momentum and number of layers were fixed. We tried different combinations of learning rate (0.01, 0.001, 0.000) and batch size (16, 32, 64). The standard error of the mean of the classification accuracies is reported.

B. Experiment 1: AID ↔ UCM

In order to assess the performance of the proposed framework for the AID ↔ UCM experiment, the 13 common classes of the AID/UCM datasets were used: agriculture, airplane, baseball, beach, dense residential, forest, medium residential, parking, playground, port, river, sparse residential, and storage (as shown in Fig. 5). For the UCM dataset, each class has 100 images with a size of 256 × 256 pixels and a spatial resolution of 1 ft. For the AID dataset, each class of the datasets contains 220–420 images with a size of 600 × 600 pixels and a spatial resolution ranging from 0.5 to 0.8 m.

Tables I and II list the best results obtained when transferring from AID to UCM and when transferring from AID to

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TABLE III

OVERALL CLASSIFICATION ACCURACY (%) COMPARISON FOR THE NWPU ↔ UCM EXPERIMENT USING VGG16 AS THE BACKBONE NETWORK

Method	N_U(%)	U_N(%)
Source only	78.07±1.34	58.42±0.58
DDC	74.10±2.03	55.77±0.62
DAN	88.14±0.43	62.72±1.55
Deep CORAL	90.30±0.52	58.27±0.62
DANN	87.70±0.61	62.38±0.51
DATSNET_L1	89.86±0.42	73.62±0.51

TABLE IV

OVERALL CLASSIFICATION ACCURACY (%) COMPARISON FOR THE NWPU ↔ UCM EXPERIMENT USING RESNET-50 AS THE BACKBONE NETWORK

Method	N_U(%)	U_N(%)
Source only	70.20±0.52	61.55±0.42
DDC	80.47±0.65	52.12±0.31
DAN	85.27±1.63	54.15±2.26
Deep CORAL	79.75±0.43	52.21±0.29
DANN	80.40±0.33	53.31±0.22
DATSNET_L1	91.87±0.25	77.53±1.22

UCM. A_U denotes transferring from AID to UCM, and U_A denotes transferring from UCM to AID.

When transferring from AID to UCM, as can be seen in Table I, the cross-scene classification result of DATSNET_L1 with a batch size of 64 and a learning rate of 0.001, i.e., 81.44%, is much better than that of the DDC, Deep CORAL, and DANN methods when using VGG16. A similar result is obtained when using ResNet-50 as the backbone network.

When transferring from UCM to AID, as shown in Tables I and II, the cross-scene classification result of DATSNET_L1 with a batch size of 64 and a learning rate of 0.001, i.e., 77.12%, is much better than that of the DDC, DAN, Deep CORAL, AANN, and DANN methods when using VGG16. Similar results are also obtained when using ResNet-50 as the backbone network. It can thus be said that the proposed method performs well when conducting domain adaptation from a dataset with a small amount of samples to a dataset with a large amount of samples.

In Tables I and II, on transferring from AID to UCM using ResNet-50 as the backbone network, DATSNET_L1 still shows a robust performance in the different tasks and with different network backbones.

C. Experiment 2: NWPU ↔ UCM

In order to assess the performance of the proposed method for the NWPU ↔ UCM experiment, the 20 common classes of the NWPU/UCM datasets were used: airport, baseball, beach, bridge, church, commercial, dense residential, desert, farmland, forest, industrial, meadow, medium residential, mountain, parking lot, railway station, river, sparse residential, and storage tanks (as shown in Fig. 6). The purpose of this experiment was to test the performance with datasets with a greater contrast.

Tables III and IV list the results obtained when transferring from NWPU to UCM and when transferring from UCM to NWPU. N_U denotes transferring from NWPU to UCM, and U_N denotes transferring from UCM to NWPU.

Table III lists the results of the proposed method and the other cross-scene classification methods of DDC, DAN, Deep CORAL, and DANN using VGG16 as the backbone network in the NWPU ↔ UCM experiment. Table III lists the results of the proposed method and the other cross-scene classification methods using ResNet-50 as the backbone network.

TABLE V

OVERALL CLASSIFICATION ACCURACY (%) COMPARISON FOR THE NWPU ↔ AID EXPERIMENT USING VGG16 AS THE BACKBONE NETWORK

Method	N_A(%)	A_N(%)
Source only	85.50±0.21	65.57±0.57
DDC	82.10±0.79	82.55±0.36
DAN	92.14±0.26	86.77±0.32
Deep CORAL	92.49±0.52	85.18±0.63
DANN	91.72±0.63	85.72±0.51
DATSNET_L1	92.88±0.12	86.06±0.35

For transferring from NWPU to UCM, as can be seen in Table III, the cross-scene classification result of DATSNET_L1 with a batch size of 64 and a learning rate of 0.001, i.e., 89.86%, is much better than that of the DDC, DAN, Deep CORAL, and DANN methods. A similar result is obtained when using ResNet-50 as the backbone network.

For transferring from UCM to NWPU, as can be seen in Table III, the cross-scene classification result of DATSNET_L1 with a batch size of 64 and a learning rate of 0.001, i.e., 73.62%, is much better than that of the DDC, DAN, Deep CORAL, and DANN methods when using VGG16 as the backbone network. A similar result is also obtained when using ResNet-50 as the backbone network. It can therefore be said that the proposed method performs well when conducting domain adaptation from a dataset with a small amount of samples to a dataset with a large amount of samples.

In Tables III and IV, it can be seen that, although the Deep CORAL method performs well in transferring from NWPU to UCM when using VGG as the backbone network, DATSNET_L1 still shows a robust performance in the different tasks and with different network backbones.

D. Experiment 2: NWPU ↔ AID

In order to evaluate the performance of the proposed framework for the NWPU ↔ AID experiment, the 19 common classes were used: airport, baseball, bridge, beach, church, commercial, dense residential, desert, farmland, forest, industrial, meadow, medium residential, mountain, parking lot, railway station, river, sparse residential, and storage tanks (as shown in Fig. 7).

Tables V and VI list the results obtained when transferring from NWPU to AID and when transferring from AID

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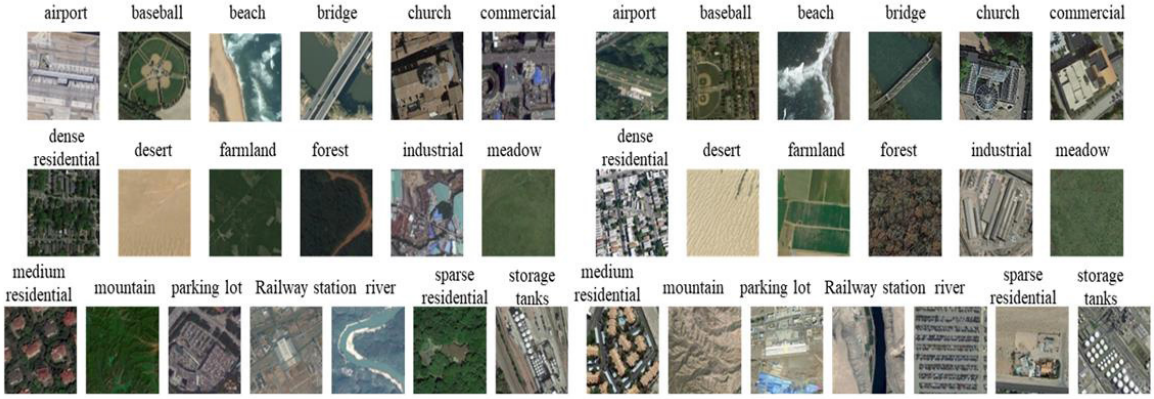


Fig. 6. Samples of the 20 common classes in the NWPU and UCM datasets.

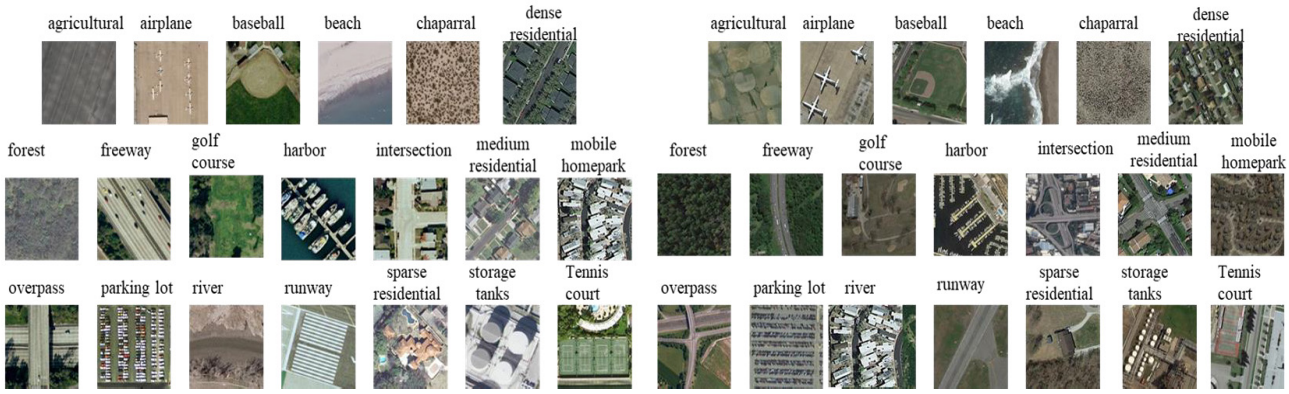


Fig. 7. Samples of the 19 common classes in the AID and NWPU datasets.

TABLE VI
OVERALL CLASSIFICATION ACCURACY (%) COMPARISON FOR THE NWPU ↔ AID EXPERIMENT USING RESNET-50 AS THE BACKBONE NETWORK

Method	N_A(%)	A_N(%)
Source only	88.21±1.45	74.53±1.32
DDC	85.30±0.31	81.31±0.44
DAN	86.66±1.28	86.72±1.32
Deep CORAL	85.10±0.35	81.40±0.30
DANN	86.08±0.28	82.27±1.37
DATSNET_L1	94.33±0.53	88.05±0.75

to NWPU. N_A denotes transferring from NWPU to AID, and A_N denotes transferring from AID to NWPU.

Table V lists the results of the proposed method and the other cross-scene classification methods of DDC, DAN, Deep CORAL, and DANN using VGG16 as the backbone network in the NWPU ↔ AID experiment. Table V lists the results of the proposed method and the other cross-scene classification methods using ResNet-50 as the backbone network. The purpose of this experiment was to test the performance when transferring from a large dataset to another large dataset.

When transferring from NWPU to AID, as can be seen in Table V, the cross-scene classification result of DATSNET_L1 with a batch size of 64 and a learning rate of 0.001, i.e., 92.88%, is much better than that of the DDC, DAN, Deep CORAL, and DANN methods when using VGG16 as the backbone network. A similar result is also obtained when using ResNet-50 as the backbone network.

When transferring from AID to NWPU, as can be seen in Table V, the cross-scene classification result of DATSNET_L1 with a batch size of 64 and a learning rate of 0.001, i.e., 88.05%, is much better than that of the DDC, DAN, Deep CORAL, and DANN methods when using VGG16 as the backbone. A similar result is also obtained when using ResNet-50 as the backbone network. It can therefore be said that the proposed method performs well when conducting domain adaptation from a dataset with a large amount of samples to another dataset also with a large amount of samples. In Tables V and VI, it can be seen that DATSNET_L1 remains robust in the different tasks and with the different network backbones.

E. Experiment 3: Influence of the Task-Specific Classifier Discrepancy

Since the task-specific classifier discrepancy used to align the feature distributions between the source domain and the target domain is a key measure in the proposed framework, we investigated the sensitivity of different task-specific classifier discrepancy measures when optimizing the networks in an adversarial manner. The measurement of the distance between the two different distributions was conducted with the Manhattan distance, the KL divergence, and the Wasserstein distance. The proposed framework was evaluated in six tasks:

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TABLE VII

COMPARISON OF THE DIFFERENT CLASSIFICATION RESULTS WHEN USING DIFFERENT DISCREPANCY MEASURES FOR TRANSFERRING FROM UCM TO AID, WITH VGG16 AS THE BACKBONE NETWORK

Method	Total	agriculture	airplane	baseball	beach	dense	forest	medium	parking	play ground	port	river	sparse	storage
DATSNET KL	0.62	0.33	0.23	0.79	0.84	0.74	0.93	0.73	0.88	0.66	0.40	0.42	0.68	0.72
DATSNET L1	0.78	0.18	0.28	0.99	0.90	0.90	0.96	0.79	0.98	0.89	0.68	0.85	0.91	0.94
DATSNET WA	0.70	0.49	0.17	0.80	0.81	0.79	0.94	0.81	0.93	0.75	0.54	0.63	0.72	0.85

from UCM to AID, from AID to UCM, from UCM to NWPU, from NWPU to UCM, from AID to NWPU, and from NWPU to AID. Tables VII–XII list the quantitative results for the six tasks when using VGG16 as the backbone network.

1) AID ↔ UCM: Tables VII and VIII, respectively, list the different classification accuracy results obtained when using the different discrepancy measures for transferring from UCM to AID and from AID to UCM, with VGG16 as the backbone network. When transferring from AID to UCM, it can be seen that DATSNET_L1 obtains a superior accuracy on most classes. It is also apparent that DATSNET_L1 obtains a better accuracy on the scenes with complicated land-use distribution spatial characteristics. When transferring from UCM to AID, DATSNET_L1 obtains a superior accuracy on most classes.

2) NWPU ↔ UCM: Tables IX and X, respectively, list the different classification accuracies obtained using the different discrepancy measures when transferring from NWPU to UCM and from NWPU to UCM, with VGG16 as the backbone network. In Table VIII, when transferring from UCM to NWPU, DATSNET_L1 obtains a superior accuracy on most classes, while DATSNET_WA obtains a better accuracy on the forest class. In Table IX, when transferring from NWPU to UCM, DATSNET_L1 obtains a superior accuracy on most classes. The experiment also shows that DATSNET_L1 can obtain a higher accuracy than DATSNET_KL and DATSNET_WA. The results indicate that DATSNET_L1 shows a robust performance when transferring from a dataset with a large amount of samples to a dataset with a small amount of samples and vice versa.

3) NWPU ↔ AID: Tables XI and XII, respectively, list the different classification results obtained using different discrepancy measures when transferring from NWPU to AID and from NWPU to AID, with VGG16 as the backbone network. In Tables VIII and IX, it can be seen that DATSNET_L1 obtains a superior accuracy on most classes, which indicates that the L1 measure is a more robust measure than the Wasserstein distance.

In Tables X and XI, it can be seen that DATSNET_L1 is robust to the different tasks and different network backbones, whereas KL divergence does not perform as well. One possible reason for this is that KL divergence is an asymmetric measure, which may not be suitable for measuring the discrepancy.

From the results, it can be seen that the KL divergence only performs well in some special cases. The Wasserstein distance is computationally intensive but, in some cases, can obtain a better performance. The L1 distance is the most robust and time-efficient measure, and we recommend that the L1 distance is used in real-world applications.



Fig. 8. Categories of the OSAR32 dataset.



Fig. 9. Original Wuhan Center Google image and the road network for Wuhan city center.

F. Experiment 3: Semantic Annotation of the Wuhan Center Google Image Dataset

To evaluate the proposed framework on a real-world application, we tested the proposed method on the Wuhan Center Google image dataset.

For the source domain, from the existing open-source OpenStreetMap (OSM) dataset [59] and the existing open-source remote sensing datasets of SIRI-WHU [7], AID [54], and NWPU-RESISC45 [55], 32 urban scene related categories were selected to form the OSAR32 dataset (as shown in Fig. 8). In total, we collected 9600 images of 256 × 256 pixels, with each category consisting of 300 images, with the spatial resolution ranging from approximately 0.2 to 30 m.

For the target domain, for urban scene annotation, how to define the boundary of a specific scene object remains an open question. The definition of a specific scene parcel in the proposed framework is a spatially continuous and functionally homogeneous area bounded by road or river. The method proposed in [60] was introduced in this experiment to generate the boundaries of the urban scene parcels. The overall workflow of generating urban scene parcels is given

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TABLE VIII

COMPARISON OF THE DIFFERENT CLASSIFICATION RESULTS WHEN USING DIFFERENT DISCREPANCY MEASURES FOR TRANSFERRING FROM AID TO UCM, WITH VGG16 AS THE BACKBONE NETWORK

Method	total	agriculture	airplane	baseball	beach	dense	forest	medium	parking	play ground	port	river	sparse	storage
DATSNET KL	0.76	0.76	0.71	0.74	0.99	0.62	0.98	0.79	0.99	0.72	0.71	0.38	0.88	0.66
DATSNET L1	0.87	0.88	0.86	0.95	1.00	0.62	1.00	0.91	1.00	0.86	1.00	0.44	0.92	0.81
DATSNET WA	0.82	0.83	0.78	0.85	0.99	0.66	0.98	0.80	1.00	0.81	0.97	0.49	0.93	0.58

TABLE IX

COMPARISON OF THE DIFFERENT CLASSIFICATION RESULTS WHEN USING DIFFERENT DISCREPANCY MEASURES FOR TRANSFERRING FROM UCM TO NWPU, WITH VGG16 AS THE BACKBONE NETWORK

Method	total	agriculture	airplane	base	beach	chaparral	dense	forest	freeway	golf course	harbor	intersection	medium	mobile	overpass	parking	river	runway	sparse	storage	tennis
DATSNET _KL	0.56	0.12	0.63	0.68	0.92	0.92	0.55	0.93	0.49	0.70	053	0.64	0.53	0.66	0.56	0.64	0.3	0.48	0.65	0.58	0.55
DATSNET _L1	0.73	0.12	0.77	0.78	0.83	0.99	0.65	0.96	0.70	0.97	0.86	0.81	0.74	0.81	0.88	0.76	0.40	0.76	0.86	0.95	0.74
DATSNET _WA	0.63	0.22	0.70	0.55	0.77	0.97	0.54	0.97	0.66	0.77	0.74	0.69	0.62	0.78	0.54	0.69	0.26	0.51	0.76	0.66	0.72

TABLE X

COMPARISON OF THE DIFFERENT CLASSIFICATION RESULTS WHEN USING DIFFERENT DISCREPANCY MEASURES FOR TRANSFERRING FROM NWPU TO UCM, WITH VGG16 AS THE BACKBONE NETWORK

Method	total	agriculture	airplane	base	beach	chaparral	dense	forest	freeway	golf course	harbor	intersection	medium	mobile	overpass	parking	river	runway	sparse	storage	tennis
DATSNET _KL	0.81	0.63	0.99	0.78	0.83	1.00	0.40	0.99	0.92	0.81	0.91	0.90	0.74	0.74	0.82	0.91	0.41	0.88	0.88	0.65	0.91.
DATSNET _L1	0.89	0.87	1.00	0.90	0.99	1.00	0.53	1.00	0.96	0.89	1.00	0.5	0.81	0.77	0.92	1.00	0.56	0.98	0.97	0.77	1.00
DATSNET _WA	0.85	0.82	1.00	0.82	0.96	1.00	0.48	0.93	0.93	0.83	0.99	0.89	0.74	0.68	0.91	0.91	0.49	0.96	0.88	0.82	0.99

TABLE XI

COMPARISON OF THE DIFFERENT CLASSIFICATION RESULTS WHEN USING DIFFERENT DISCREPANCY MEASURES FOR TRANSFERRING FROM AID TO NWPU, WITH VGG16 AS THE BASELINE NETWORK

Method	total	airport	Baseball	beach	bridge	church	commercial	dense	desert	farmland	forest	industrial	meadow	medium	mountain	Parking lot	railway station	river	sparse	storage
DATSNET _KL	0.76	0.65	0.64	0.94	0.93	0.73	0.61	0.62	0,85	0.84	0.80	0.74	0.64	0.79	0.88	0.86	0.86	0.65	0.58	0.92
DATSNET _L1	0.86	0.82	0.82	1.00	0.95	0.80	0.66	0.77	0.95	0.96	0.96	0.81	0.82	0.82	0.77	0.88	0.91	0.80	0.85	0.98
DATSNET _WA	0.82	0.67	0.76	0.95	0.97	0.81	0.58	0.79	0.94	0.93	0.93	0.85	0.78	0.68	0.84	0.91	0.91	0.71	0.73	0.94

as follows: 1) all of the existing OSM road data in the OSM data layers are extracted as line features for a new road feature layer; 2) the broken road sections are trimmed at a threshold of 200 m to remove isolated road parts; 3) manual topology correction is undertaken for the second trimmed road layer; and 4) the urban scene parcels are generated as buffer zones around the road networks, with different buffer thresholds. The original Wuhan Center Google image, as shown in Fig. 9, was collected from Google Earth in June 2019, covering the central part of the city of Wuhan in China. The spatial resolution of the image is 1 m.

To evaluate the performance of DATSNET, we labeled 1130 image patches for the target domain. The experimental results obtained with ResNet-50, DAN, Deep CORAL, and DANN are shown for comparison with the proposed method. The results of the different domain adaptation methods are listed in Table XIII. From the table, it can be seen that DATSNET_L1 achieves a better overall classification accuracy than the other state-of-the-art domain adaptation methods.

In Fig. 10, it can be seen that DATSNET_L1 performs better on the bare land, commercial, and water classes. However, the medium residential, school, and commercial classes are highly confused. The meadow, park, and greenbelt classes also show a great deal of confusion. One possible reason for this is that the appearances of the different categories at the image patch level are insufficiently different, and there is a wide variation

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TABLE XII

COMPARISON OF THE DIFFERENT CLASSIFICATION RESULTS WHEN USING DIFFERENT DISCREPANCY MEASURES FOR TRANSFERRING FROM NWPU TO AID, WITH VGG16 AS THE BASELINE NETWORK

Method	total	air	Base	beach	bridge	church	comm ercial	dense	desert	farmla nd	forest	industr ial	meado w	mediu m	mount ain	Parkin g lot	Railwa y station	river	sparse	storage
DATSNET _KL	0.84	0.84	0.99	0.87	0.82	0.84	0.85	0.83	0.97	0.81	0.95	0.68	0.82	0.84	0.94	0.99	0.88	0.67	0.74	0.89
DATSNET _L1	0.92	0.92	1.00	0.98	0.93	0.93	0.90	0.86	0.98	0.94	0.97	0.80	0.97	0.78	0.98	0.99	0.95	0.92	0.92	0.95
DATSNET _WA	0.87	0.84	0.99	0.89	0.86	0.91	0.81	0.77	0.96	086	0.95	0.75	0.94	0,79	0.91	0.99	0.92	0.74	0.86	0.86

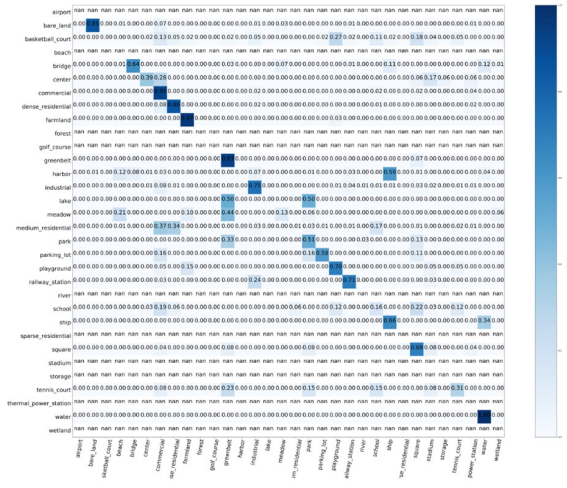


Fig. 10. Confusion matrix for the Wuhan Center Google image dataset with the proposed DATSNET framework.

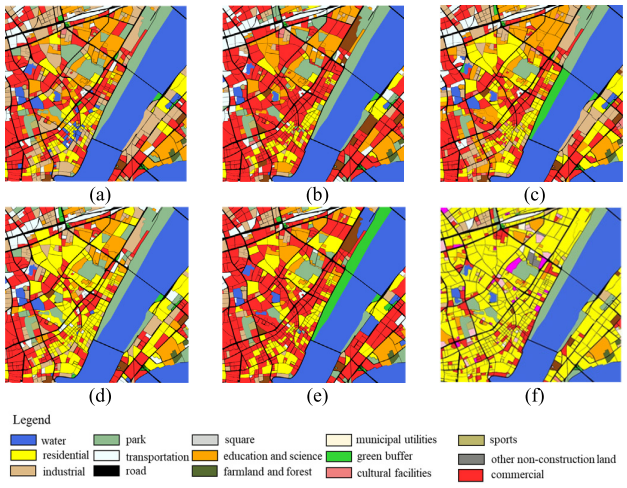


Fig. 11. Annotated image results for the Wuhan Center Google image dataset. (a) RESNET. (b) Deep CORAL. (c) DANN. (d) DAN. (e) DATSNET_L1. (f) Ground truth.

in land-use distribution and spatial scales when compared with the source domain images.

The annotation results obtained with DATSNET and the other comparison methods for the Wuhan Center Google image dataset are shown in Fig. 11.

TABLE XIII

CLASSIFICATION RESULTS FOR THE WUHAN CENTER GOOGLE IMAGE

Method	RESNE T	Deep CORAL	DANN	DAN	DATS_L1
OA	59.58	57.60	60.97	64.69	66.58

The commercial scene and the residential scene show similar spatial patterns in the textural characteristics. As a result, these two scene classes show confusion and poor classification accuracy. Some other misclassifications also occur, mainly because of the unseen scene categories and the imbalanced class distribution. Another possible reason is that the spatial distribution and the scale of the land-use objects contained in this scene are very different from the spatial distribution and the scale of the land-use objects in the open-source remote sensing scene classification datasets.

V. CONCLUSION

In this article, the DATSNET framework has been proposed for cross-scene HSR imagery scene classification. The proposed method is designed to address the problem of scene classification when transferring from existing datasets to a new dataset and is able to improve the alignment of the different classifiers' feature distributions by minimaxing the defined discrepancy, to optimize the networks in an adversarial manner. The proposed method aligns the distributions of the source domain features and target domain features by exploiting task-specific class decision boundaries. As a robust approach, DATSNET obtained superior performances in the cross-scene experiments conducted in this study, even when transferring from the existing datasets to an unseen area. The urban scene label transfer experiment undertaken in this study also showed that the proposed DATSNET framework achieved an acceptable performance in the urban scene label transfer task. The proposed method is better able to discover the common knowledge between the source domain and the target domain than the conventional domain adaptation methods, especially when utilizing the task-specific decision boundaries in the target domain.

The alignment of the different classifiers' feature distributions still remains an open problem for cross-scene HSR imagery scene classification. In our future work, we will continue exploring how to better align the different classifiers' feature distributions based on the analysis of multisource datasets.

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For real-world applications, the amounts of different scenes in a city may follow a long-tail distribution, which can harm the scene classification accuracy. We will focus on the negative effect of the long-tail distribution of the target domain images in a specific city in our future work.

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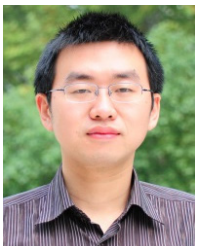
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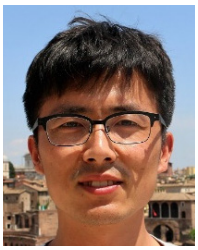
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